

Tackling Group Theory computationally with GAP

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GAP as a Programming Language

Looking at Groups in GAP

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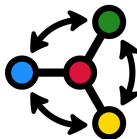
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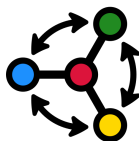
History

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- ▶ GAP was started at Lehrstuhl D für Mathematik, RWTH Aachen in 1986.



RWTHAACHEN
UNIVERSITY

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- ▶ It has a yearly conference (This year in Portugal)
- ▶ It also has an yearly excellence awards!

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- ▶ Provide a rich library of functions for algebraic computations.
- ▶ Support for programming and scripting to automate tasks.
- ▶ Extensive documentation (gap-system.org/doc) and a supportive community.

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- ▶ Number of groups can get BIG (non-isomorphic groups of order ≤ 2000 (423 million groups!)).
- ▶ Can and has helped uncover patterns not easily seen at first.

What we're going to do today?

We're going to try to solve three group theory problems using GAP:

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2. Determine if the automorphism group of a non-abelian group is also non-abelian.
3. Conjecture trends about the commutativity ratio.

Online setup

We will be using mybinder to run GAP in an online Jupyter notebook. For a local setup, please refer to the GAP installation page, but we warn this might take time.

mybinder

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- ▶ This makes it ideal for interactive exploration and rapid prototyping
- ▶ You can type commands directly and see results immediately

Basic Syntax

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# This is a comment
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y := x + 3;;      # No output
x = 5;           # Returns true
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Lists are fundamental in GAP:

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- ▶ Access elements with `list[i]`
- ▶ List ranges: `[1..10]` creates `[1, 2, ..., 10]`

```
mylist := [10, 20, 30];  
mylist[1];          # Returns 10  
mylist[2];          # Returns 20  
primes := [2, 3, 5, 7, 11];
```

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- ▶ `if ... then ... elif ... else ... fi`
- ▶ `for ... in ... do ... od`
- ▶ `while ... do ... od`
- ▶ `function ... end`

```
for i in [1..5] do
  if i mod 2 = 0 then
    Print(i, " is even\n");
  fi;
od;
```

Functions

Defining functions in GAP:

```
square := function(x)
    return x * x;
end;
```

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square(5);  # Returns 25
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```

Anonymous functions with `->`:

```
double := x -> 2 * x;
double(7);  # Returns 14
```

Your turn

Write a function that computes the n -th Fibonacci number.

Recall: The Fibonacci sequence is defined as:

- ▶ $F(0) = 0$
- ▶ $F(1) = 1$
- ▶ $F(n) = F(n-1) + F(n-2)$ for $n \geq 2$

Test your function with: `Fib(10)`

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Presentation of group

GAP provides built-in functions for a lot of known group types.
These include:

- ▶ $\mathbb{Z}_n$ as `CyclicGroup(n)`
- ▶ S_n as `SymmetricGroup(n)`
- ▶ A_n as `AlternatingGroup(n)`
- ▶ D_n as `DihedralGroup(2n)`
- ▶ Matrix Groups
- ▶ Quaternion Groups

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G := Group((1,2,3)(4,3,5));
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G := Group((1,2,3)(4,3,5));
```

Furthermore, any arbitrary group's elements will be represented by permutations (why?).

Analyzing a group

GAP can answer problems about the group and its elements. It can:

- ▶ List all the elements - `Elements(G)`
- ▶ List all the subgroups - `AllSubgroups(G)`
- ▶ List all the normal subgroups - `NormalSubgroups(G)`
- ▶ Get the order of an element - `Order(g)`

Analyzing a group

GAP can also check if a certain property holds for a group by prepending the property with the `Is` noun:

- ▶ `IsAbelian`
- ▶ `IsCyclic`
- ▶ `IsNormal`
- ▶ `IsIsomorphic`

Your turn

Choose some group and find the following properties:

- ▶ Order
- ▶ Number of Conjugacy classes
- ▶ Order of the Largest Proper Normal Subgroup
- ▶ Center
- ▶ Exponent (google what that is)
- ▶ Minimal Generating set

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Testing Conjectures

GAP provides an easy setting to check statements and conjectures. In this section, we want to check whether the following is true or false.

Is the automorphism group of a non-abelian group non-abelian?

Morphisms

In GAP, a morphism is a function between two objects of the same type (category).

$$\text{Set} \rightarrow \text{Set}$$
$$\text{Grp} \rightarrow \text{Grp}$$
$$\text{Ring} \rightarrow \text{Ring}$$

Group Homomorphisms

For groups, we can define a homomorphism in two ways:

1. Defining the image of the generators.
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1. Defining the image of the generators.
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We also have `IsHomomorphism` to check if a function actually can be a homomorphism.

Group Automorphism

Automorphisms for groups follow the same rules as homomorphisms.

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We also have `AutomorphismGroup(G)` to get the automorphism group.

Your turn

Check our claim again.

Is the automorphism group of a non-abelian group non-abelian?

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Finding Trends

GAP can also be great at analyzing trends and ideas.

Commutativity Ratio

For a finite group G the probability that two group elements chosen at random with replacement commute, which we denote by $\text{Pr}(G)$, is given by

$$\text{Pr}(G) = \frac{|\{(a, b) \in G \times G \mid ab = ba\}|}{|G|^2}$$

We call $\text{Pr}(G)$ the *commutativity ratio* of G .

Your turn

- ▶ Compute $\text{Pr}(G)$ for dihedral groups of order 6, 8, and 12
- ▶ Look for patterns in the results
- ▶ Can you conjecture a formula for $\text{Pr}(G)$ in terms of the number of conjugacy classes?

Your turn

- ▶ Compute $\text{Pr}(G)$ for dihedral groups of order 6, 8, and 12
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Hint: The number of conjugacy classes can be found using `NrConjugacyClasses(G)`.

Searching All Groups

GAP maintains a library of all groups up to certain orders. You can search through them using:

- ▶ `AllSmallGroups( $n$ )` - Returns all groups of order n
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You can also filter by properties:

- ▶ `AllSmallGroups(Size, n, IsAbelian, true)`
- ▶ `AllSmallGroups(Size, n, IsCyclic, false)`

Your turn

Using the small groups library, find:

- ▶ How many non-abelian groups of order 12 exist?
- ▶ Which group of order 8 is the quaternion group?
- ▶ Find a non-cyclic group of order 15 (if it exists)

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Characters

GAP provides powerful tools for character theory:

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Character tables are useful for:

- ▶ Studying group representations
- ▶ Determining if groups are isomorphic
- ▶ Understanding conjugacy classes

Categories

GAP uses categories to organize algebraic structures:

- ▶ `CategoryCollections(IsObject)` - Collections of objects
- ▶ `IsMagma` - Sets with binary operations
- ▶ `IsSemigroup`, `IsMonoid`, `IsGroup` - Group-like structures
- ▶ `IsRing`, `IsField` - Ring and field structures

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You can check what category an object belongs to using the `Is` functions.

Free Presentations

GAP allows you to work with groups via presentations:

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- ▶ `/ [relator1, relator2, ...]` - Quotient by relations

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Example: Define $D_8 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$

```
F := FreeGroup("r", "s");  
r := F.1; s := F.2;  
D8 := F / [r^4, s^2, s*r*s*r];
```

GAP in Research

GAP has been instrumental in many groundbreaking mathematical discoveries:

- ▶ **Ore Conjecture (2010):** Proved that every element of every finite non-abelian simple group is a commutator. GAP was used extensively for computational verification across thousands of groups.

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- ▶ Classification of groups of small order
- ▶ Character theory computations for sporadic simple groups
- ▶ Verification of algebraic structures in representation theory

Over 3000+ papers cite GAP in their research!

Further Exploration

Want to practice more with GAP?

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- ▶ Your CURIOSITY!

Thank You!

Questions?

Resources: gap-system.org